Credit Risk Analyzer

ML Lending System

Satyam Kumar (2401010428) Krishna Verma (2401010240)
Aditya Kishore Singh (2401010031) Akshit Vats (2401020085)

Section: [**A**]

April 19, 2026

Abstract

This report outlines the development of an end-to-end machine learning system designed to predict credit card default risk. The system provides explainable AI insights via SHAP and serves real-time decisions through a production-grade Streamlit dashboard. It is built to mirror how fintech lending teams actually evaluate borrowers.

1 Problem Statement

Consumer credit default costs financial institutions billions annually. Traditional rule-based underwriting often misses complex, non-linear patterns in borrower behavior—patterns that machine learning captures effectively. This project builds a complete credit risk pipeline: from raw transaction data to a deployable web interface that a lending analyst could use to evaluate a client in under 30 seconds, with full model transparency and explainability.

2 Features

The primary features of this project include:

- **Full ML Pipeline:** Data ingestion, feature engineering, preprocessing, model training, evaluation, and serving.
- **Explainability:** SHAP-based feature importance analysis explaining the reasoning behind each prediction.
- **Production UI:** A dark-themed, responsive fintech dashboard built with Streamlit.
- **Engineered Features:** Eight derived financial indicators such as credit utilization, payment ratio, and delay severity.
- **Data-Driven Insights:** Automated risk factor analysis explaining each prediction with color-coded reasoning.

3 Tech Stack

The technology stack utilized in this project is summarized in Table 1.

Layer	**Technology**
Language	Python 3.12
ML Framework	scikit-learn
Explainability	SHAP
Web UI	Streamlit
Data	Pandas, NumPy
Serialization	Joblib
Dataset	UCI Taiwan Credit Card Default (30,000 records)

Table 1: Project Technology Stack

4 System Design & Architecture

The system architecture consists of several key layers:

1. **Data Layer:** The UCI Dataset is loaded, preprocessed, and scaled.

2. **ML Pipeline:** Feature engineering is applied, followed by model training using Logistic Regression. The model is then evaluated, and SHAP explainability is integrated.

3. **Serving Layer:** A Streamlit Dashboard accepts user input, transforms the features using a saved scaler, runs inference using the saved model, and outputs a risk score and insights.

4. **Persisted Artifacts:** Pre-trained models (`best_model.pkl`) and preprocessors (`scaler.pkl`) are saved for real-time inference.

5 Machine Learning Pipeline

The data undergoes the following transformations:

- **Raw Data:** 30,000 records with 24 features.
- **Feature Engineering:** Added features like Average Bill Amount, Credit Utility, Average Payment Delay, Payment-to-Bill ratio, Maximum Payment Delay, Number of Late Months, Payment Standard Deviation, and a Severe Delay Flag.
- **Preprocessing:** One-Hot Encoding and Standard Scaling.
- **Model Training:** Logistic Regression selected via cross-validation.
- **Evaluation & Explainability:** ROC-AUC, Confusion Matrix, and SHAP Global/Local feature importance.

6 Model Performance

The system uses a Logistic Regression model trained on 30,000 records, achieving a ROC-AUC score of approximately 0.75. This score indicates the model correctly ranks a random defaulter above a random non-defaulter 75% of the time. While it may not reach a production threshold for autonomous decisions, it provides strong directional guidance for human-in-the-loop lending workflows.

7 Milestone 2: Agentic AI Underwriting Copilot

Building upon the core machine learning pipeline, Milestone 2 expands the project into an Agentic AI solution representing modern enterprise AI architecture. This introduces an Automated Underwriting Copilot with the following capabilities:

- **Automated OCR Extraction Pipeline:** Utilizes Large Language Models (Groq Llama 3.3) and PDF parsing to automatically extract unstructured financial data from raw bank statements and map them perfectly to the 24 required model features.

- **Human-in-the-Loop Orchestration:** A LangChain ReAct agent orchestrates prediction routing, enforcing an escalation workflow where "Grey Zone" risk predictions (e.g., 30%–60% likelihood of default) halt automated processing and ping a human loan officer for manual review.
- **Context-Aware Intelligence:** Loan officers can interact with the Copilot through a chat interface to ask dynamic follow-up questions regarding the applicant's raw data and the SHAP explainability report.

8 Future Improvements

Future iterations of this project could focus on:

- **Threshold Optimization:** Tuning the decision threshold using the precision-recall tradeoff for business-specific cost matrices.
- **Model Upgrades:** Experimenting with XGBoost, LightGBM, and neural networks for improved AUC.
- **Cloud Deployment:** Containerizing with Docker and deploying to Streamlit Community Cloud or Render.
- **API Layer:** Creating a FastAPI endpoint for programmatic integration.
- **Monitoring:** Adding prediction drift detection and model performance tracking.

9 Conclusion

The Credit Risk Analyzer successfully demonstrates an end-to-end machine learning system tailored for the financial sector. By combining robust feature engineering, reliable modeling, responsive Agentic AI orchestration, and transparent explanations, it provides actionable insights in a format suitable for rapid review by lending professionals.

10 Team Contribution

The project was developed collaboratively with clearly defined responsibilities:

- **Aditya Kishore Singh:** Conducted research and prepared the research documentation, including literature review and theoretical foundation.
- **Krishna Verma:** Developed the Streamlit frontend dashboard and designed the project presentation (PPT).
- **Satyam Kumar:** Built the complete machine learning pipeline including preprocessing, feature engineering, model training, evaluation, SHAP integration, the Agentic AI Copilot orchestration, and system deployment.
- **Akshit Vats:** Managed the dataset, performed exploratory data analysis, and ensured data quality for modeling.